Time as an "embodied" component of meaning

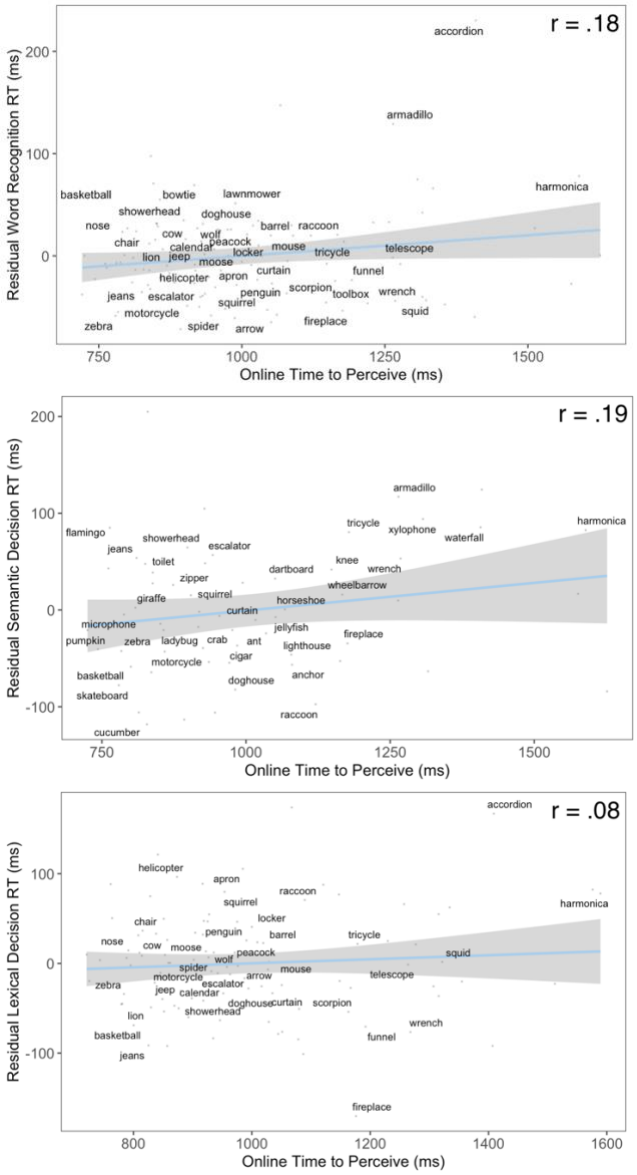
Introduction. Our conceptual knowledge (e.g., our knowledge of what a *funnel* or a *zebra* is) determines how we comprehend language. But what information does conceptual knowledge include? According to experience-based (e.g., embodied) theories, any systematic components of our experiences with things could be included,[1] and indeed, studies suggest that sensorimotor, emotional, and interoceptive experiences (among others) are all part of conceptual knowledge[2]. However, temporal experience—specifically, how long it takes to perceptually apprehend an instance of something—has received relatively little attention.

Prior work on sentence processing shows that people simulate the temporal unfolding of described events: sentences describing ongoing, temporally extended events (e.g., "we were approaching the summit") elicit slower reading times than sentences describing punctive events, and these reading times correlate with people's estimates of how long those events would take to unfold.[3] This suggests that temporal information is simulated during sentence processing. But what about the individual concepts that make up events—are their temporal properties also encoded and simulated during word processing?

There is some evidence that they are: people's ratings of how long it takes to perceive an instance of a concept (its *time-to-perceive*) predict how long it takes them to recognize the corresponding word, even after controlling for a wide range of lexical and semantic variables.[4] This pattern is consistent with the hypothesis that temporal aspects of real-world experience become part of lexical representations and are reactivated during word recognition. However, this interpretation hinges on the assumption that subjective time-to-perceive ratings correspond to actual perceptual experience. Here we (1) directly test that assumption and (2) ask whether a perceptual "ground truth" measure of *time-to-perceive* predicts "time-to-conceive," as measured by response times in lexical decision, word recognition, and semantic decision tasks.[5-7]

Methods. To obtain a ground truth measure of time-to-perceive, participants (N=100) were sequentially presented with 147 images of objects, and pressed the space bar as soon as they recognized the object (they were told that 'recognizing' meant knowing what the object was, as opposed to recalling its name, allowing us to measure image recognition time independently of naming time). To compare these image recognition times to subjective ratings, we asked a separate group of participants to rate English words referring to those 147 objects on *time-to-perceive* ('How much time would it take to accumulate the information needed to perceive X?'; N=44). We also collected ratings from separate groups of participants on two control variables: (1) *space-to-perceive* ('How much space would it take to accumulate the information needed to perceive X?'; N=*24*), and (2) *confusability* ('How difficult would it be to distinguish the following thing from similar things?'; N=35).

Results. *Time-to-perceive* ratings strongly predicted image recognition times even after controlling for word length, frequency, age of acquisition, concreteness, visual perceptual strength, object agreement, space-to-perceive and confusability ($p < .001$). This suggests that people's intuitions about *time-to-perceive* indeed reflect perceptual experience. More importantly, using the same control variables, image recognition time predicted our metric of "time-to-*conceive*", i.e., response times, in megastudies of word recognition ($p < .01$; top panel) and semantic decision ($p < .05$; middle panel), although not in the (shallower) lexical decision task ($p = .29$; bottom panel). These results extend temporal-simulation effects from sentences to words: just as comprehenders simulate the temporal unfolding of events during sentence processing, they appear to simulate the amount of time it takes to perceptually apprehend the referent of a word during word processing. These findings support the idea that how long it takes to perceive something is part of conceptual knowledge, and that accessing a word involves implicitly embodying or simulating this time—such that concepts that require more time to perceive take longer to mentally "play out" during word processing.



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